English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Mathematics / Applied Mathematics, 24 June 2024 Examination *The grade is 2 + the number of points. Time allowed: 4 hours. Good luck! We use n, k for positive integers and t for a positive real number.*

Problem 1. 1. (0.5 points) Let Ω have 12 elements. In how many ways can we choose three subsets $A, B, C \subset \Omega$ satisfying

$$\mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) = 1 + \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C)?$$

2. (0.5 points) Let Ω be the set of nondecreasing functions from $\{1, \dots, k\}$ to $\{1, \dots, n\}$. What is the probability that a uniformly chosen element of Ω is strictly increasing?

Problem 2. A student passes the time tossing a coin and keeps track of the excess of heads over tails. Let S_n be this excess after n tosses.

1. (0.25 points) Since the coin is fair, the student recalls that the law of large numbers implies $\lim_{n\to\infty} \mathbb{P}(S_n = 0) = 1$. Explain why this is or is not correct.
2. (0.5 points) Find the limits, as $n \to \infty$, of $\mathbb{P}(S_{2n} = 0)/n$, $\mathbb{P}(S_{2n} = 0)$, $n\mathbb{P}(S_{2n} = 0)$ and $\sqrt{n}\mathbb{P}(S_{2n} = 0)$.[1]
3. (0.5 points) Approximate $\mathbb{P}(S_n \in (0, \epsilon\sqrt{n}))$ for large n and small ϵ.

Problem 3. Let $X_1, X_2, \dots$ be independent Exp(1) variables and $S_n = X_1 + \cdots + X_n$.

1. (0.5 points) Find the density of S_2 and prove by induction that $f_{S_n}(x) = x^{n-1}e^{-x}\mathbb{1}_{\{x>0\}}/(n-1)!$.
2. (0.25 points) Let $N_t = \max\{n : S_n \le t\}$. Prove that $\mathbb{P}(N_t \ge n) = \mathbb{P}(S_n \le t)$ and $\mathbb{P}(N_t = n) = \mathbb{P}(S_n \le t) - \mathbb{P}(S_{n+1} \le t)$.
3. (0.25 points) Find the distribution of N_t.

Problem 4. Benford's law,[2] also known as the first-digit law, is the empirical observation that in many real datasets the first nonzero digit (not only in decimal notation) is not uniform and is most often 1. The decimal Benford distribution assigns probability $\log_{10}(1 + 1/d)$ to digit $d \in \{1, \dots, 9\}$.

Find, to accuracy 0.01, the probability that the first nonzero decimal digit of X is 1 when:

1. (0.25 points) $X \sim N(0, 1)$;
2. (0.25 points) $X \sim \text{Exp}(1)$;
3. (0.5 points) $X = U_1/U_2$ for independent $U_1, U_2 \sim U(0, 1)$. Also find $\mathbb{E}X$.
4. (Bonus, 0.25 points) X is a "random positive power of 2". Formally, find $\lim_{n\to\infty} \mathbb{P}(X_n$ begins with 1) for $X_n = 2^{U_n}$, $U_n \sim U(\{1, \dots, n\})$.

Compare with Benford's distribution.[3]

Problem 5. Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, and suppose a sample of n observations is available.

1. (0.25 points) Derive maximum likelihood estimators of μ, σ^2.
2. (0.25 points) Construct 95% confidence intervals for μ and σ.
3. (0.5 points) Test $H_0 : \mu = 1$ against $H_1 : \mu > 1$ at level $\alpha = 0.05$. Can H_0 be rejected for $n = 100$, $\overline{X} = 1.1$, $S^2 = 1.2213$?

[1] You may use Stirling's formula $n! \sim \sqrt{2\pi n}(n/e)^n$.
[2] Observed by Newcomb in logarithm tables in 1881, and by Benford in a larger dataset in 1934.
[3] If time permits, calculate the probabilities for digits greater than 1 and compare them with Benford's distribution.